

# Soft sensor: separator vapor fraction & product composition

---

**Goal.** Continuously infer a gas-oil separator's vapor fraction (a GOR-like proxy) and gas/liquid product compositions from cheap, continuous measurements (temperature, pressure), instead of requiring a continuous composition analyzer on every separator.

A production/gas-oil separator splits a wellstream into gas and liquid at its operating temperature and pressure. Given the feed composition, the split (vapor fraction) and both product compositions are fully determined by vapor-liquid equilibrium — a PT flash. In the field, separator T and P are cheap, continuous, reliable measurements; feed (wellstream) composition usually is **not** continuously measured — it comes from an infrequent well test or lab GC sample. This notebook's soft sensor evaluates the flash continuously from live T/P, using the last known feed composition, so it tracks changes in operating conditions between well tests instead of going stale.

**Measurements assumed available:** feed composition (z, from a periodic well test / lab GC — not continuous), separator temperature (T) and pressure (P), both continuous. **Inferred / computed by this soft sensor:** vapor fraction, gas composition, liquid composition. **Tier:** A — direct PT-flash calculation (docs/measurement\_tiers.md), given that *some* feed composition is known (even if only periodically). If feed composition is not available at all, the flash is underdetermined and this drops to Tier C — you'd need a broader well/reservoir model or historical correlation to fill in the missing boundary condition.

This report uses [Clapeyron.jl](#)'s Peng-Robinson EOS and multicomponent PT-flash solver for a representative wellstream (methane / ethane / propane / n-butane / n-pentane / n-hexane), evaluated at a fixed operating point (the interactive notebook version lets you adjust feed composition, temperature and pressure with sliders).

## 2. Operating point used in this printout

Feed composition (normalized): [0.45, 0.15, 0.15, 0.1, 0.08, 0.07]

Separator temperature: **40.0 °C**, pressure: **20.0 bar**

```
components = ["methane", "ethane", "propane", "butane", "pentane", "hexane"]  
model = PR(components)
```

### 3. Soft sensor output: PT flash (vapor fraction & product composition)

**Vapor fraction (soft-sensor output): 71.8 mol%** of the feed leaves as gas at 40.0 °C / 20.0 bar.

| Component | Feed z | Gas y | Liquid x |
|-----------|--------|-------|----------|
| Methane   | 0.45   | 0.601 | 0.065    |
| Ethane    | 0.15   | 0.176 | 0.083    |
| Propane   | 0.15   | 0.137 | 0.184    |
| n-Butane  | 0.1    | 0.056 | 0.213    |
| n-Pentane | 0.08   | 0.022 | 0.228    |
| n-Hexane  | 0.07   | 0.008 | 0.227    |

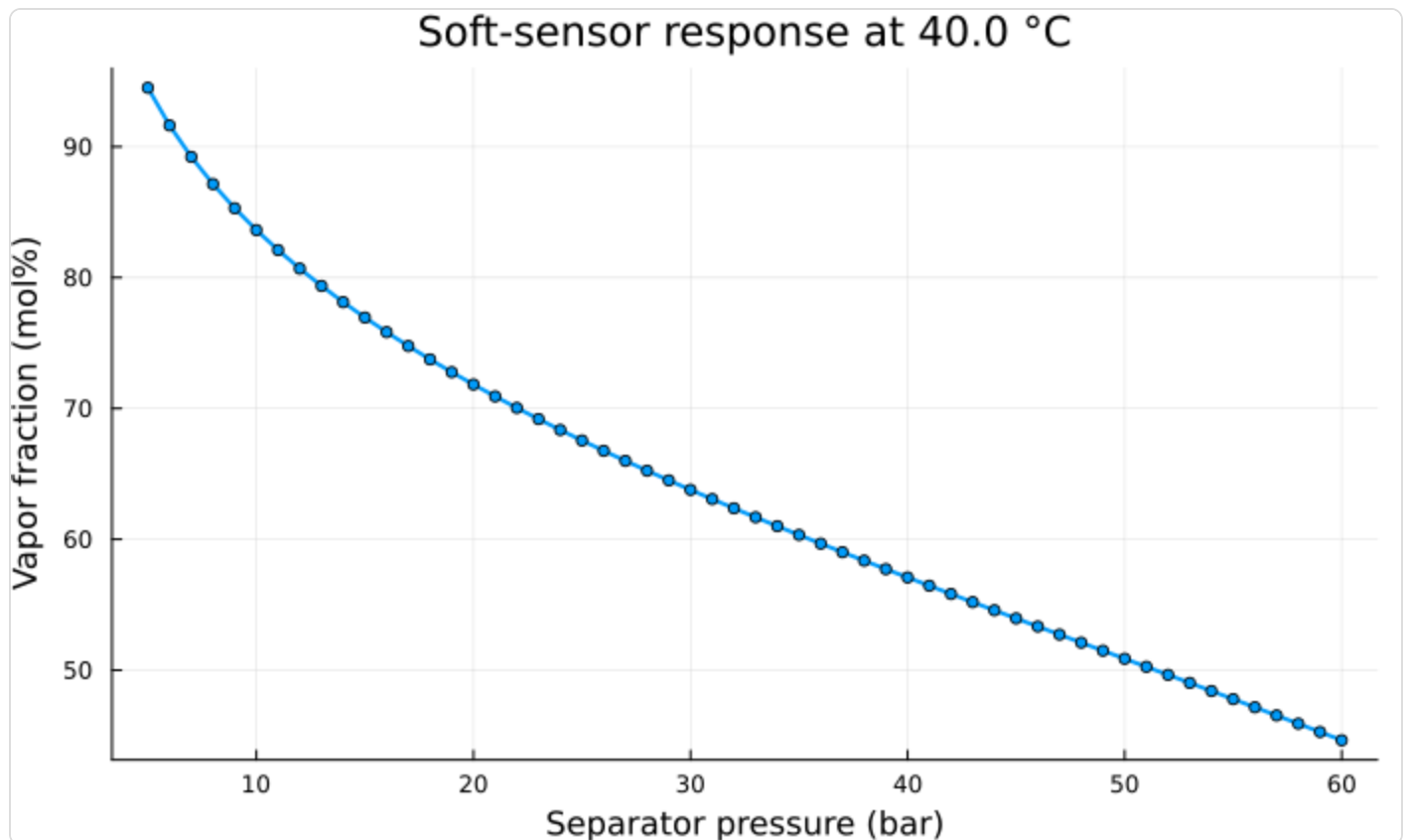
In a real deployment,  $z$  would be refreshed from each well test / lab sample while this calculation re-evaluates continuously from the live T/P scan, so gas and liquid product quality track operating changes between tests.

```
xphase, nphase, G = tp_flash(model, Psep, Tsep, z, RRTPFflash())  
vapor_fraction = sum(nphase[vap_idx, :]) / sum(nphase)
```

Vapor fraction  $\approx$  71.8 mol% at 40.0 °C / 20.0 bar

### 4. Sensitivity: vapor fraction vs separator pressure

Sweep separator pressure at fixed temperature and feed composition — the classic separator-pressure-optimization curve. A soft sensor should move smoothly and monotonically here; this is a basic sanity check before trusting it against plant data.



## 5. Next steps toward a deployable soft sensor

- Feed this soft sensor from the plant historian: live separator T/P every scan, feed composition  $z$  refreshed on each well test / lab GC sample (held constant between updates).
- Validate vapor fraction and gas/liquid compositions against any available continuous gas flow meter or periodic lab analysis; add a bias/trim term if a consistent offset appears.
- Track how much accuracy degrades between well tests as the real wellstream composition drifts — this bounds how long a Tier A treatment stays valid before it should be re-classified as Tier B/C (composition effectively unmeasured) and paired with a well/reservoir decline model.
- Extend to three-phase flash (add free water) for wellstreams with produced water, and to slug/transient behavior if the separator sees significant flow variability.